

```
from google.colab import files
uploaded = files.upload()
```

Choose Files Dataset.csv

Dataset .csv(text/csv) - 2249716 bytes, last modified: 2/23/2026 - 100% done

Saving Dataset .csv to Dataset .csv

```
df = pd.read_csv("Dataset .csv")
```

```
df.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Current
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botswai Pula(
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Botswai Pula(
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Botswai Pula(
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.056475	14.585318	Japanese, Sushi	...	Botswai Pula(
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.057508	14.584450	Japanese, Korean	...	Botswai Pula(

5 rows × 21 columns

```
df = df[['Restaurant Name', 'Cuisines', 'City', 'Price range', 'Aggregate rating', 'Votes']]
df.head()
```

	Restaurant Name	Cuisines	City	Price range	Aggregate rating	Votes	
0	Le Petit Souffle	French, Japanese, Desserts	Makati City	3	4.8	314	
1	Izakaya Kikufuji	Japanese	Makati City	3	4.5	591	
2	Heat - Edsa Shangri-La	Seafood, Asian, Filipino, Indian	Mandaluyong City	4	4.4	270	
3	Ooma	Japanese, Sushi	Mandaluyong City	4	4.9	365	
4	Sambo Kojin	Japanese, Korean	Mandaluyong City	4	4.8	229	

Next steps:

Generate code with df

New interactive sheet

```
df['Cuisines'] = df['Cuisines'].fillna('Unknown')
df['City'] = df['City'].fillna('Unknown')
```

```
encoded_features = pd.get_dummies(df[['Cuisines', 'City', 'Price range']])
```

```
numeric_features = df[['Aggregate rating', 'Votes']]

feature_matrix = pd.concat([encoded_features, numeric_features], axis=1)
feature_matrix.head()
```

	Price range	Cuisines_Afghani	Cuisines_Afghani, Mughlai, Chinese	Cuisines_Afghani, North Indian	Cuisines_Afghani, North Indian, Pakistani, Arabian	Cuisines_African	Cuisines_African, Portuguese	Cuisines_American
0	3	False	False	False	False	False	False	False
1	3	False	False	False	False	False	False	False
2	4	False	False	False	False	False	False	False
3	4	False	False	False	False	False	False	False
4	4	False	False	False	False	False	False	False

5 rows × 1970 columns

```
similarity_matrix = cosine_similarity(feature_matrix)
```

```
def recommend_restaurants(cuisine, city, price_range, top_n=5):

    # Filter restaurants matching user preference
    filtered_df = df[
        (df['Cuisines'].str.contains(cuisine, case=False)) &
        (df['City'].str.lower() == city.lower()) &
        (df['Price range'] == price_range)
    ]

    if filtered_df.empty:
        return "No matching restaurants found."

    # Get index of first matching restaurant
    idx = filtered_df.index[0]

    # Get similarity scores
    sim_scores = list(enumerate(similarity_matrix[idx]))

    # Sort by similarity
    sim_scores = sorted(sim_scores, key=lambda x: x[1], reverse=True)

    # Get top recommendations (excluding itself)
    sim_scores = sim_scores[1:top_n+1]

    restaurant_indices = [i[0] for i in sim_scores]

    return df.iloc[restaurant_indices][['Restaurant Name', 'Cuisines', 'City', 'Aggregate rating']]
```

```
recommend_restaurants("Italian", "New Delhi", 2)
```

	Restaurant Name	Cuisines	City	Aggregate rating	
5857	Chicago Pizza	Fast Food, Italian, Pizza	New Delhi	3.4	
3012	MOB Brewpub	Continental, Italian, Asian, Indian	New Delhi	4.7	
3258	Cafe Yell	Cafe, Continental, Italian	New Delhi	4.4	
2572	Nariyal Cafe	Continental, Seafood, Goan, Andhra, Kerala, Thai	New Delhi	4.2	
6844	Cafe Hashtag LoL	Cafe, North Indian, Chinese	New Delhi	4.2	

